

# The intraganglionic loop: sympathetic–sensory coupling in the dorsal root ganglion as a mechanism for the sensory symptoms of hard flaccid syndrome

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The underlying conjecture is credited to the hard flaccid syndrome community member writing as *Literature Greedy*; see Acknowledgements.

Third in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*.

## Abstract

**Background.** Hard flaccid syndrome (HFS) is usually framed as a smooth muscle disorder, yet its sensory features are among its most consistent: numbness in 74.8% of surveyed patients, alongside dysesthesia, paresthesia and pain, and worsening with masturbation in 75.9%. No account of the smooth muscle state explains why sensory symptoms should track sympathetic arousal so closely.

**Hypothesis.** We propose that after the initiating nerve injury, sympathetic postganglionic axons sprout within the dorsal root ganglia serving the penis and form functional coupling with sensory neuron somata, so that sympathetic activity — which rises on standing, with stress and with sexual arousal — directly drives sensory discharge. Through the somato-sympathetic reflex, that discharge may in turn raise sympathetic outflow, closing a loop inside the ganglion.

**Basis.** Sympathetic sprouting into the dorsal root ganglion after peripheral nerve injury is a well-replicated animal finding, and sympathetic stimulation can then excite the encircled neurons. Basket-like sympathetic skeins have been observed in dorsal root ganglia removed from human neuropathic pain patients. Injured sensory neurons up-regulate  $\alpha 2$ -adrenoceptors, and in a subset of patients with neuropathic pain, cutaneous noradrenaline reproduces or aggravates the pain.

**What the evidence does not support.** Ultrastructural work shows that sympathetic sprouts do not make direct contacts with the sensory soma membrane; coupling must occur by diffusion of transmitter through extracellular space. Sprouting occurs preferentially around large-diameter neurons rather than nociceptors. In at least one rat study, the magnitude of sprouting in the injured sacral ganglion did not differ between animals with well-developed and poorly-developed neuropathic pain. Systemic  $\alpha$ -adrenergic blockade has been argued not to identify a distinct sympathetically maintained pain population at all, and only 7 of 25 patients classified as having sympathetically maintained pain responded to cutaneous noradrenaline in the largest such study.

**Predictions.** The hypothesis makes one prediction that no other account of HFS makes: sympathetic blockade should relieve dysesthesia and pain *even in patients whose hard flaccid state does not change*. We set out a two-by-two outcome grid in which each quadrant supports a different hypothesis. We also propose a cutaneous noradrenaline provocation test applied to penile and perineal skin, adapted from an established human protocol — a cheap, non-invasive, blindable test that has never been performed in this condition — and identify a receptor-subtype prediction that the commonly prescribed  $\alpha 1$ -selective antagonists would not be expected to satisfy.

**Conclusion.** The hypothesis rests on a real and replicated phenomenon, explains features that nothing else explains, and is testable without new technology. It is also weaker than it first appears: the sprouting literature contains substantial negative findings, the loop closure is entirely unsupported, and the one clinical datum usually cited against it tested the wrong ganglionic level and is uninformative.

**Keywords:** hard flaccid syndrome; dorsal root ganglion; sympathetic sprouting; sympathetically maintained pain;  $\alpha 2$ -adrenoceptor; volume transmission; genito-pelvic dysesthesia.

## Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. The provocation test and blocks described are research procedures requiring formal ethical approval; intradermal catecholamine injection into genital skin has never been performed and its safety in that territory is unknown. Nothing here should influence clinical management.

## 1. Introduction

Hard flaccid syndrome (HFS) is named for a smooth muscle sign — a persistently semi-rigid flaccid penis — and the mechanistic literature has followed the name. The prevailing account holds that a pathological somato-visceral reflex drives excess sympathetic efferent activity that contracts cavernosal smooth muscle [5, 6].

The symptom data do not fit that emphasis comfortably. In the largest survey to date ( $n = 143$ , self-selected online sample), numbness or loss of sensation somewhere on the penis was reported by 74.8% of respondents, and the condition worsened with masturbation in 75.9% [B] [3]. Pain, dysesthesia and paresthesia are described consistently across the case literature [1, 4, 2], and expert consensus places HFS within the persistent genital arousal disorder / genito-pelvic dysesthesia spectrum — a sensory classification [B] [7]. A theory of cavernosal smooth muscle does not explain any of this, and in particular does not explain why sensory symptoms should track sympathetic arousal.

That coupling is the puzzle this paper addresses. Sympathetic outflow rises on standing, with psychological stress, and substantially during sexual arousal and orgasm. If sensory symptoms rise with it, the simplest explanation is that sympathetic activity is reaching the sensory apparatus directly — and there is a well-characterised, if contested, mechanism by which exactly that can happen after nerve injury: sympathetic postganglionic axons sprout within the dorsal root ganglion (DRG) and form functional coupling with sensory neuron somata [12].

This paper states that hypothesis for HFS, grades the evidence in both directions, and specifies how it could be tested. It is the third in a series examining conjectures from a source article on HFS mechanism [9]; the underlying idea in this case is not ours and is credited in the Acknowledgements.

### 1.1 Relationship to the companion papers

The first companion paper asks which anatomical route carries pathological sympathetic traffic to the penis [10]; the second asks what generates that traffic [11]. The present paper asks a different question again: what the traffic does to the *sensory* system on its way past. The three are compatible and partly interlocking — the loop proposed here, if real, would be one of the mechanisms sustaining the central state proposed in the second paper.

## 2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *sympathetic sprouting dorsal root ganglion*, *sympathetic-sensory coupling*, *sympathetically maintained pain*, *noradrenaline-evoked pain*, *alpha-2 adrenoceptor sensory neuron*, *dorsal root ganglion stimulation*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic; no protocol was registered and no formal risk-of-bias appraisal was applied. We searched deliberately for disconfirming as well as confirming evidence, and Section 5 reports what that search returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

**Table 1.** Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

| Grade | Definition                                                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------------|
| [A]   | Replicated human evidence: consistent findings across independent human studies.                                       |
| [B]   | Single human study, human reference text, congress abstract, or human data with substantial methodological limitation. |
| [C]   | Experimental animal evidence, with species and preparation stated.                                                     |
| [D]   | Mechanistic inference: deduced from A–C evidence but not itself directly observed.                                     |
| [E]   | Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.                             |

## 3. Background

### 3.1 Sympathetic sprouting into the dorsal root ganglion

After sciatic nerve ligation in the rat, noradrenergic perivascular axons sprout into the dorsal root ganglion and form basket-like structures around the somata of large-diameter axotomised sensory neurons; sympathetic stimulation can then activate those neurons repetitively [C] [12]. The finding has been replicated in several injury models and laboratories: rapid sprouting after chronic constriction injury [C] [VERIFY] [16], varicosity formation after spinal nerve injury [C]

[[VERIFY](#)] [15], and dependence on satellite-glia-derived nerve growth factor and neurotrophin-3 [[C](#)] [[VERIFY](#)] [17].

Functionally, nerve injury triggers sympathetic–sensory coupling within rat DRGs; both excitatory and suppressive responses occur, and the latency between onset of sympathetic stimulation and the excitatory response in A-fibre afferents averaged roughly 10 seconds, with suppressive responses averaging roughly 20 seconds [[C](#)] [13]. Those latencies matter and we return to them.

The substrate on the sensory side also changes. Injured or inflamed primary afferent neurons up-regulate  $\alpha 2$ -adrenoceptors [[C](#)] [[VERIFY](#)] [18], and ectopic  $\alpha 2$ -adrenoceptors in axotomised sensory neurons couple to N-type calcium channels [[C](#)] [[VERIFY](#)] [19]. Many DRG somata express  $\alpha 2A$ -adrenoceptors both intracytoplasmically and on the membrane surface [[C](#)] [14].

### 3.2 The coupling is by diffusion, not synapse

This is the most important mechanistic detail in the literature and it is routinely glossed over. Ultrastructural work in rat found that sympathetic sprouts grow on the surface of glial lamellae that form around the neuron, but that *sympathetic axons never make direct contacts with the soma membrane*; the authors concluded that the functional coupling known to occur must be mediated by diffusion of transmitter molecules through the extracellular space [[C](#)] [14].

Three consequences follow, and each generates a testable expectation [[D](#)]:

1. **Slow.** Volume transmission over a diffusion distance produces responses on a timescale of seconds to tens of seconds, which is what was observed [13]. Any account of HFS symptoms invoking this mechanism should predict symptom changes over seconds to minutes after a sympathoexcitatory stimulus, not milliseconds.
2. **Spatially imprecise.** Transmitter diffusing in extracellular space will reach neurons that were never encircled, including neurons serving neighbouring territories. This predicts symptom spread beyond the injured nerve's distribution.
3. **Pharmacologically accessible.** A diffusion-mediated signal acting at  $\alpha$ -adrenoceptors on the sensory soma should be interruptible by an adrenoceptor antagonist reaching the ganglion, and the relevant subtype is more likely  $\alpha 2$  than  $\alpha 1$  [14, 18]. This has a direct and awkward clinical implication, developed in Section 8.4.

### 3.3 Human evidence

The critical human finding is that sympathetic basket-skeins have been observed in dorsal root ganglia removed from human neuropathic pain patients — though the same authors were explicit that a functional relation between those structures and the patients' symptoms remained unestablished [[B](#)] [14]. Pericellular nests were described in sensory ganglia by Cajal long before the modern literature, and have been reported in healthy animals as well, which raises the possibility that some such structures are a normal feature rather than an injury response [[C](#)] [[VERIFY](#)] [23].

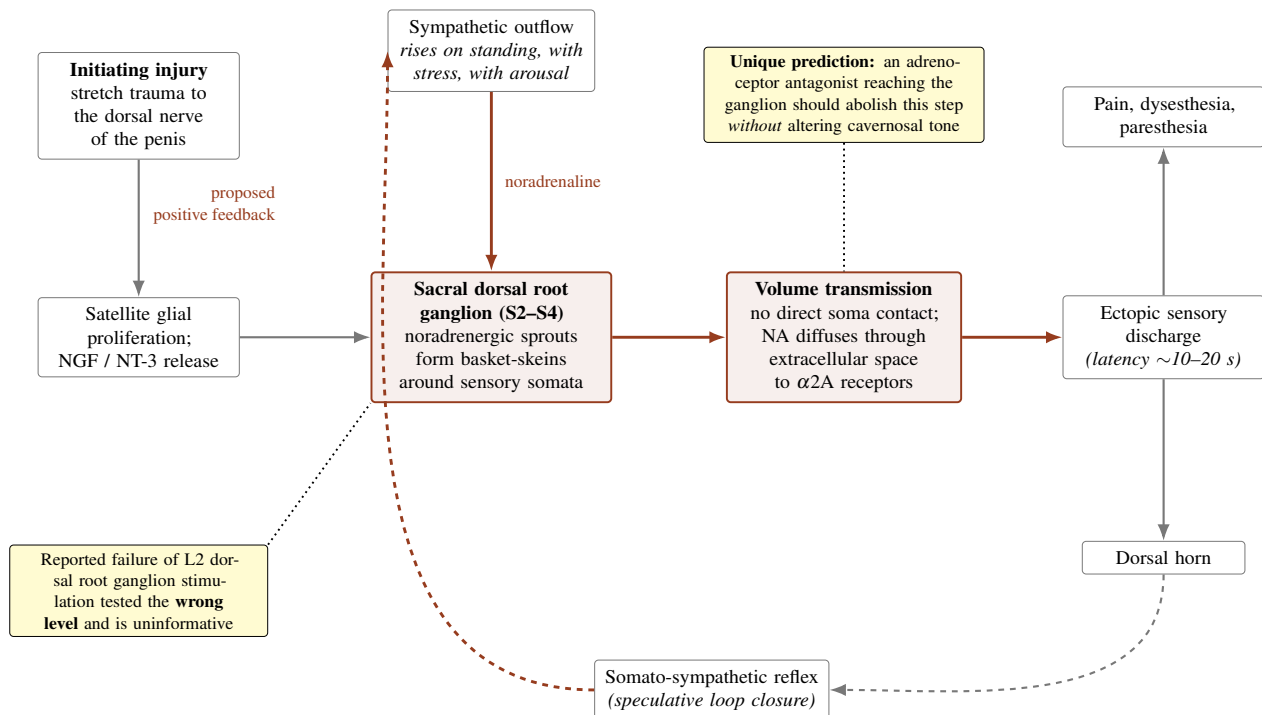
Human pharmacological evidence comes from the sympathetically maintained pain literature. Cutaneous application of noradrenaline to symptomatic skin aggravated pain and mechanical or thermal hyperalgesia in 7 of 25 patients classified as having sympathetically maintained pain, and in none of 10 classified as sympathetically independent [[B](#)] [25]. Intradermal noradrenaline evoked pain in patients with sympathetically maintained pain [[B](#)] [[VERIFY](#)] [26]. In complex regional pain syndrome, experimental modulation of sympathetic vasoconstrictor activity was related to pain and hyperalgesia in a case–control study [[B](#)] [[VERIFY](#)] [27], and pain increased during sympathetic arousal [[B](#)] [[VERIFY](#)] [28].

We emphasise the denominator in the first of those: **7 of 25**, in patients pre-selected for sympathetic dependence [25]. Even in the population where the phenomenon should be easiest to demonstrate, it was demonstrable in a minority.

### 3.4 Which ganglia serve the penis

The hypothesis is level-specific, and the levels are not the ones usually assumed. Somatic afferents from the penile skin and glans travel in the dorsal nerve of the penis and the pudendal nerve to the S2–S4 dorsal root ganglia. Visceral afferents from the erectile tissue travel with the sympathetic supply and reach dorsal root ganglia at approximately T10–L2. Distinct electrophysiological subclasses of DRG neuron innervating the glans penis and distal urethra have been characterised [[C](#)] [[VERIFY](#)] [32]. Sympathetic fibres are found in the pudendal nerve as well as the pelvic and cavernous nerves [[B](#)] [34].

There are therefore *two* candidate ganglionic populations, at widely separated segmental levels, with different sympathetic sources: sacral ganglia receiving postganglionic fibres from the sacral sympathetic trunk, and thoracolumbar ganglia receiving them locally. The hypothesis should specify which. We propose the sacral ganglia as the primary site, because the sensory symptoms of HFS map to the somatic territory of the dorsal penile nerve rather than to a visceral referral pattern [[D](#)].



**Figure 1. The proposed intraganglionic loop.** Injury to the dorsal nerve of the penis is proposed to trigger satellite glial proliferation and noradrenergic sprouting within the sacral dorsal root ganglia, forming basket-like skeins around sensory somata. Sympathetic activity — which rises on standing, with stress and with sexual arousal — then releases noradrenaline that reaches up-regulated  $\alpha 2A$ -adrenoceptors on the sensory soma by diffusion rather than through a synapse, evoking ectopic discharge with a latency of seconds to tens of seconds. The resulting afferent volley produces the sensory symptoms directly and, through the somato-sympathetic reflex, is proposed to raise sympathetic outflow further, closing a loop inside the ganglion. **The dashed loop closure is the weakest element of the hypothesis and has no supporting evidence in any species.** Diagram is schematic and not to anatomical scale.

## 4. The hypothesis

### Formal statement

**H3 (intraganglionic sympathetic–sensory coupling).** After the initiating injury in hard flaccid syndrome, sympathetic postganglionic axons sprout within the dorsal root ganglia serving the penis — principally the sacral ganglia — and form functional coupling with sensory neuron somata. Sympathetic activity therefore drives sensory discharge directly, producing pain, dysesthesia and paresthesia that vary with sympathetic tone.

**H3-open (the defensible form).** The coupling is a one-way efferent-to-afferent link. It explains the sensory symptoms and their dependence on posture, stress and sexual arousal. It makes no claim about the hard flaccid state.

**H3-closed (the ambitious form).** The evoked sensory discharge feeds the somato-sympathetic reflex, raising sympathetic outflow further, so that the loop is self-reinforcing and contributes to maintaining the smooth muscle state as well.

**H3-null.** Sensory symptoms in HFS arise from partial deafferentation and central changes, with no sympathetic–sensory coupling at the ganglion.

The distinction between the open and closed forms is not cosmetic. **H3-open is supported by an established phenomenon and explains something no other conjecture explains. H3-closed requires an additional step for which there is no evidence in any species.** We recommend that the hypothesis be advanced, tested and defended in its open form, and that the loop closure be treated as a separate and much more speculative proposal. Conflating them is the main way this hypothesis could be oversold.

Auxiliary assumptions:

- **A1 (ganglionic specificity).** Sprouting occurs in the sacral dorsal root ganglia after a distal penile nerve injury. Never demonstrated; sprouting has been shown after sciatic, spinal nerve and caudal trunk injuries but *not* in the trigeminal ganglion after inferior alveolar or infraorbital nerve injury [C] [VERIFY] [22], so it is not a universal ganglionic response [DATA NEEDED].
- **A2 (injury severity).** A stretch injury sustained during masturbation is sufficient to trigger sprouting. The animal

literature uses ligation, transection and chronic constriction — injuries of a different order [DATA NEEDED].

- **A3 (nociceptor involvement).** The coupling reaches neurons that signal pain and dysesthesia. Sprouting is preferentially around *large-diameter* neurons [12, 14]; see Section 5.

## 5. Evidence against, stated first

We put the disconfirming evidence before the confirming evidence deliberately, because this hypothesis is unusually easy to present persuasively and the negative literature is substantial.

1. **Sprouting magnitude does not track pain.** In rats subjected to unilateral transection of the caudal trunk between the S1 and S2 spinal nerves, the magnitude of tyrosine-hydroxylase-positive fibre penetration into the injured S1 dorsal root ganglion did not differ significantly between animals with well-developed neuropathic pain behaviours and animals with poorly-developed ones; the authors concluded that sprouting in the injured DRG is not a key factor in the development of neuropathic pain [C] [VERIFY] [20]. This study is directly relevant twice over: it is a *sacral* ganglion, and it is a dissociation experiment of exactly the kind H3 needs to survive.
2. **The wrong cells.** Baskets form preferentially around large-diameter neurons [12, 14] — classically low-threshold mechanoreceptive afferents, not nociceptors. Some later work reports that basket-encircled cells show increased nociceptive markers and are a major source of spontaneous activity [C] [VERIFY] [24], but the original anatomical association is with the wrong population for a pain hypothesis.
3. **Adrenergic blockade does not reliably reverse the behaviour.** Mechanical hyperalgesia after spinal nerve ligation in rats was not reversed by intraplantar or systemic adrenergic antagonists [C] [VERIFY] [21].
4. **The clinical construct itself is contested.** Phentolamine blockade has been argued not to identify a distinct sympathetically maintained pain population, questioning the concept [B] [VERIFY] [29]. Even in the affirmative literature the response rate is a minority: 7 of 25 [25].
5. **Sympathectomy has weak clinical support.** A Cochrane review of cervico-thoracic or lumbar sympathectomy for neuropathic pain and complex regional pain syndrome found the evidence insufficient to support its use [A] [VERIFY] [30]. If interrupting sympathetic outflow reliably relieved sympathetically coupled sensory symptoms, that literature would look different.
6. **Not all ganglia sprout.** No signs of sympathetic sprouting were found in the trigeminal ganglion after inferior alveolar or infraorbital nerve injury in rats [C] [VERIFY] [22]. Whether sacral ganglia behave like lumbar ones or like trigeminal ones is unknown.
7. **Pericellular nests may be normal.** Structures resembling the sprouted baskets were described in sensory ganglia of healthy animals [C] [VERIFY] [23], raising the possibility that at least some reported sprouting reflects normal anatomy detected more readily after injury-induced changes in staining or glial reaction.

Taken together, this is not a knock-down case against H3, but it is enough that the hypothesis should not be presented as resting on settled science. The correct summary is: *the phenomenon is real and replicated; its causal contribution to pain is disputed; and it has never been looked for in the ganglia at issue here.*

## 6. Evidence for, and what it does and does not show

### 6.1 What H3 explains that nothing else does

Three observations are accounted for by H3-open and by no other conjecture in this series.

**Worsening with masturbation (75.9%) [3].** Sexual arousal and orgasm involve substantial sympathetic activation. Under a smooth muscle account, masturbation should if anything relax the erectile tissue. Under H3, a sympathetic surge directly drives a sensitised sensory apparatus, so symptoms worsen [D].

**The tight coupling of a smooth muscle sign to sensory complaints.** HFS presents as a package of tone and sensation. H3 supplies a single anatomical locus at which sympathetic activity produces both.

**The dissociation between sensory and tonic responses to treatment.** If a sympathetic block relieves dysesthesia while leaving the hard flaccid state unchanged, no smooth muscle hypothesis predicts it and H3-open does. This is developed in Section 7.

### 6.2 The datum usually cited against it is uninformative

Dorsal root ganglion stimulation at L2 is reported to have failed in a single patient [E] [9], and this has been treated as evidence against the conjecture. It is not, for two independent reasons. First, one patient at one level is not a test. Second

**Table 2.** Evidence bearing on the intraganglionic coupling hypothesis, graded.

| Finding                                                                                                                                                                            | Grade        | What it establishes — and what it does not                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noradrenergic axons sprout into the DRG after sciatic ligation and form baskets around large-diameter axotomised neurons; sympathetic stimulation activates them repetitively [12] | [C]          | Establishes the phenomenon and its functional consequence in rat. Does not establish it in sacral ganglia, after stretch injury, or in humans.        |
| Nerve injury triggers sympathetic–sensory coupling within rat DRGs; excitatory latency ~10 s [13]                                                                                  | [C]          | Establishes functional coupling and its slow timescale. Does not establish that the coupled neurons signal pain.                                      |
| Basket-skeins observed in DRGs removed from human neuropathic pain patients [14]                                                                                                   | [B]          | Establishes that the structure occurs in humans. The authors explicitly did not establish a functional relation to symptoms.                          |
| Sympathetic sprouts make no direct soma contacts; coupling must occur by diffusion [14]                                                                                            | [C]          | Refines the mechanism and predicts slow, spatially imprecise, pharmacologically accessible coupling. Weakens any claim of a tight, synapse-like link. |
| Injured sensory neurons up-regulate $\alpha 2$ -adrenoceptors; ectopic $\alpha 2$ receptors couple to N-type calcium channels [18, 19]                                             | [C]          | Supplies the receptor substrate on the sensory side. Subtype specificity in humans not established.                                                   |
| Cutaneous noradrenaline aggravated pain in 7/25 patients with sympathetically maintained pain, 0/10 sympathetically independent [25]                                               | [B]          | Establishes that adrenergic sensitivity of afferents occurs in some patients. The minority response rate is the headline.                             |
| Pain increases during sympathetic arousal in CRPS [28]                                                                                                                             | [B] [VERIFY] | Consistent with the symptom pattern in HFS. Does not localise the coupling to the ganglion.                                                           |
| Somato-sympathetic reflexes are established and spinally organised [31]                                                                                                            | [C]          | Supplies the return limb the closed form requires. Does not show that ganglionic coupling engages it.                                                 |

and more importantly, **the ganglia serving penile cutaneous sensation are sacral, not L2** (Section 3.4). A negative result at L2 is a test of the wrong level. Human dorsal root ganglion stimulation has been shown to reduce sympathetic outflow and blood pressure [B] [VERIFY] [33], so the modality is capable of engaging the relevant physiology — but only if applied where the relevant ganglia are.

## 7. The discriminating prediction

H3-open makes a prediction that separates it cleanly from the smooth muscle conjectures: **sympathetic blockade should relieve sensory symptoms even in patients whose hard flaccid state does not change**. Existing reports of block outcomes record only whether the hard flaccid state changed [9]. The sensory outcome was not recorded. It is free to record.

|                            | Hard flaccid state unchanged                                                                                                                                                                                                                           | Hard flaccid state improved                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensory symptoms relieved  | <p><b>Sensory relieved, tone unchanged</b></p> <p><i>Supports H3-open.</i></p> <p>Sympathetic activity reaches the sensory apparatus; the smooth muscle state is maintained by something else (effector autonomy, or the wrong route was blocked).</p> | <p><b>Both relieved</b></p> <p><i>Supports a common sympathetic driver.</i></p> <p>Consistent with H3-closed, and equally with a purely efferent account in which one block interrupts both outputs. Does not discriminate.</p>                          |
| Sensory symptoms unchanged | <p><b>Neither relieved</b></p> <p><i>Supports effector autonomy, a peripheral generator, or technical failure of the block.</i></p> <p>Uninterpretable unless block adequacy was documented.</p>                                                       | <p><b>Tone relieved, sensory unchanged</b></p> <p><i>Argues against H3.</i></p> <p>Sympathetic traffic was interrupted and reached the effector, but the sensory symptoms are generated elsewhere — centrally, or by a peripheral ectopic generator.</p> |

**Figure 2.** The two-by-two outcome grid for a technically confirmed sympathetic block with separately recorded sensory and tonic endpoints. Only the upper-left quadrant is uniquely predicted by the intraganglionic coupling hypothesis; the lower-right quadrant argues against it; the other two do not discriminate. Existing block reports in hard flaccid syndrome record only the horizontal axis, which is why they cannot evaluate this hypothesis at all. Recording the vertical axis requires nothing but a validated sensory instrument and the decision to use one.

## 8. Proposed tests

### 8.1 Primary test: cutaneous noradrenaline provocation of penile and perineal skin

An established human protocol exists and has never been applied to this condition. Iontophoretic application or intradermal injection of noradrenaline into symptomatic skin, with pain and hyperalgesia measured before and after, distinguished responders from non-responders in patients with neuropathic pain [25, 26].

**Prediction P1.** Compared with matched controls and with a vehicle-treated contralateral or adjacent site, application of noradrenaline to symptomatic penile or perineal skin in men with HFS will evoke or aggravate pain, dysesthesia or mechanical hyperalgesia.

Design features that should be specified in advance: a vehicle-controlled, within-subject, site-randomised design with blinded assessment; a matched control group; graded doses to establish a concentration–response relationship rather than a single-dose yes/no; and a pre-specified responder definition. Expected effect size is unknown; in the closest comparable study the response rate in a pre-selected population was 7/25 [25], so a study powered on an assumption of majority response would be badly designed [DATA NEEDED].

Safety and feasibility caveats must be stated plainly: intradermal catecholamine injection has not to our knowledge been performed in genital skin, the vasoconstrictive risk in that territory has not been characterised, and iontophoresis is the more conservative delivery route. This would require careful ethical review and dose-finding, and we describe it as a research proposal only.

**Why this is the right primary test.** It is non-invasive or minimally invasive, uses an existing protocol, is properly blindable, and — crucially — does not require an objective measure of cavernosal tone, which does not exist [10, 11]. It tests the sensory limb of the hypothesis directly, on its own terms.

### 8.2 Secondary test: sympathoexcitation challenge with blinded sensory outcome

**Prediction P2.** A standardised sympathoexcitatory manoeuvre — cold pressor test, mental arithmetic, or controlled orthostatic challenge — will increase sensory symptom scores in men with HFS more than in matched controls, with an onset latency of seconds to minutes consistent with volume transmission (Section 3.2).

The obvious confound is expectancy: patients know they are being stressed. Mitigations include using a sham manoeuvre of comparable discomfort but lower sympathetic activation, using a sensory threshold measure rather than a symptom rating as the primary outcome, and documenting the sympathetic response independently rather than assuming it.

### 8.3 The dissociation test

**Prediction P3.** In a technically confirmed sympathetic block, sensory symptoms improve while the hard flaccid state does not (upper-left quadrant of Figure 2). This requires only that future block studies record a validated sensory outcome alongside the tonic one, and that block adequacy be documented — the same methodological reform recommended in both companion papers.

### 8.4 A receptor-subtype prediction with an awkward implication

If coupling is mediated by noradrenaline diffusing to  $\alpha 2A$ -adrenoceptors on the sensory soma [14, 18, 19], then the adrenoceptor subtype that matters for the *sensory* symptoms is  $\alpha 2$ , whereas the subtype that matters for *cavernosal contraction* is predominantly  $\alpha 1$  [D] [35].

**Prediction P4.**  $\alpha 1$ -selective antagonists — the agents most commonly prescribed in this population — would not be expected to relieve the ganglionic component of the sensory symptoms, even at doses that affect smooth muscle tone. Non-selective  $\alpha$ -blockade would be a better probe of the sensory limb.

We flag two serious complications rather than presenting this as a clean experiment. First,  $\alpha 2$  antagonists such as yohimbine block presynaptic autoreceptors and therefore *increase* noradrenaline release, so systemic  $\alpha 2$  blockade is not a clean way to interrupt an  $\alpha 2$ -mediated postsynaptic effect and could plausibly worsen symptoms. Second, the receptor subtype mediating sympathetic–sensory coupling remains contested in the animal literature, with  $\alpha 1$  and  $\alpha 2$  subtypes both implicated in different preparations [C] [VERIFY] [36, 37]. This prediction is therefore a direction for pharmacological investigation, not a testable claim in its current form.

### 8.5 The tissue test, and why it will not happen

The decisive anatomical test is examination of sacral dorsal root ganglia from men with HFS for perivascular noradrenergic basket formations. Sacral DRG tissue can only be obtained post-mortem or at surgery that would not be indicated in this condition, in a young population with a non-fatal disorder. We note the test for completeness and regard it as effectively unavailable [DATA NEEDED].

**Table 3.** Predictions of H3 compared with the principal alternatives.

| Observation                                                     | H3 (ganglionic coupling) | Central sensitization      | Peripheral ectopic generator                       | Discriminating value             |
|-----------------------------------------------------------------|--------------------------|----------------------------|----------------------------------------------------|----------------------------------|
| Cutaneous noradrenaline evokes or aggravates symptoms           | <b>Predicted</b>         | Not predicted              | Possible (adrenergic sensitivity of injured axons) | Moderate–high                    |
| Sensory symptoms worsen with sympathoexcitation                 | <b>Predicted</b>         | Not specifically predicted | Not specifically predicted                         | Moderate                         |
| Sympathetic block relieves sensory symptoms but not tone        | <b>Predicted</b>         | Not predicted              | Not predicted                                      | <b>High</b>                      |
| Symptoms extend beyond the dorsal penile nerve territory        | Predicted (diffusion)    | Predicted                  | Not predicted                                      | Low between the first two        |
| Symptom latency of seconds to minutes after a sympathetic surge | <b>Predicted</b>         | Not specified              | Not specified                                      | Moderate                         |
| Basket formations in sacral DRG tissue                          | <b>Predicted</b>         | Not predicted              | Not predicted                                      | <b>Decisive but unobtainable</b> |
| $\alpha$ 1-selective blockade fails to relieve sensory symptoms | Predicted                | Not specified              | Not specified                                      | Low (confounded)                 |

## 9. Alternative explanations

1. **Partial deafferentation with central change.** Reduced afferent input from a partially injured dorsal penile nerve, plus central hyperexcitability, accounts for the coexistence of numbness and dysesthesia without any ganglionic coupling [9]. This is H3’s principal competitor for the sensory findings, and it has the advantage of explaining the numbness — which H3 does not address at all.
2. **A peripheral ectopic generator.** Sodium channel remodelling in injured and neighbouring afferents produces stimulus-independent discharge [9]. Injured axons also acquire adrenergic sensitivity, so this alternative can absorb the noradrenaline provocation result if it is positive — meaning P1 supports a class of hypotheses rather than H3 specifically.
3. **Coupling at the periphery rather than the ganglion.** Sympathetic sprouting also occurs in skin after partial nerve injury [C] [VERIFY] [38]. If sympathetic–sensory interaction in HFS occurs in penile skin rather than the ganglion, every prediction above except the tissue test is unchanged, and the hypothesis loses its distinctive anatomical claim.
4. **Attentional and affective amplification.** Depression or anxiety was reported in 75% of the largest series [B] [8, 2]. Sympathetic arousal and anxious arousal are not separable by self-report, so a symptom that worsens “with stress” is compatible with either. Blinded threshold measures rather than symptom ratings are the only way to separate them.
5. **The observations may not need explaining.** Worsening with masturbation could reflect mechanical irritation of an injured nerve rather than sympathetic activation. This mundane alternative deserves explicit testing and is easily addressed by comparing symptom change after orgasm achieved with and without direct penile stimulation [DATA NEEDED].

## 10. Implications

No clinical implications are drawn. Three research implications follow.

**First, record the sensory outcome.** Every future report of a sympathetic block, nerve block or neuromodulation procedure in HFS should record a validated sensory outcome separately from the hard flaccid state. This costs nothing and is the only way the grid in Figure 2 can ever be populated.

**Second, target the right level.** Any future test of dorsal root ganglion stimulation or blockade motivated by this hypothesis should be directed at sacral rather than lumbar ganglia, and should state the level tested.

**Third, separate the two forms of the hypothesis.** H3-open should be tested on its own terms as an account of sensory symptoms. The closed-loop form should be labelled as a distinct and unsupported extension, and not used to claim explanatory credit for the smooth muscle state.

## 11. Limitations

1. **No evidence in HFS.** Sympathetic–sensory coupling has never been sought, let alone demonstrated, in this condition.
2. **Wrong ganglia, wrong injury.** The animal literature concerns lumbar and caudal ganglia after ligation, transection or chronic constriction. The hypothesis requires sacral ganglia after a stretch injury. Neither has been studied, and at least one ganglion (trigeminal) does not sprout at all.
3. **The loop closure is unsupported.** H3-closed has no evidence in any species and should not be defended.

4. **Substantial negative literature.** Section 5 lists seven distinct disconfirming findings, including a sacral-ganglion dissociation study and a Cochrane review of sympathectomy.
5. **The mechanism is weaker than usually described.** Coupling is diffusion-mediated with latencies of tens of seconds, not synaptic transmission.
6. **Cell-type mismatch.** Baskets form preferentially around large-diameter neurons, not nociceptors.
7. **H3 does not explain numbness,** which is the single most commonly reported sensory symptom [3]. A complete sensory account of HFS requires a deafferentation component that this hypothesis does not supply.
8. **The primary test is not specific.** A positive noradrenaline provocation result would support a family of adrenergic-sensitivity hypotheses, not H3 uniquely.
9. **Non-systematic search,** with several citations verified only against secondary sources; all are marked [VERIFY].
10. **Confirmation risk.** This paper was written to develop a specific conjecture. We have tried to counter it by placing the disconfirming evidence first, by separating the defensible from the ambitious form, and by identifying the mundane alternative to the masturbation observation. The reader should still discount accordingly.

## 12. Refutation criteria

- **Substantially refuted.** Graded cutaneous noradrenaline applied to symptomatic penile or perineal skin, vehicle-controlled and blindly assessed, produces no aggravation of pain, dysesthesia or hyperalgesia relative to matched controls.
- **Substantially refuted.** A technically confirmed sympathetic block relieves the hard flaccid state while leaving sensory symptoms unchanged (lower-right quadrant of Figure 2).
- **Weakened.** Sensory symptoms do not increase with a documented sympathoexcitatory challenge, or increase with a latency inconsistent with volume transmission.
- **Weakened.** Sacral dorsal root ganglia are shown, in an appropriate animal model of penile stretch injury, not to develop noradrenergic sprouting.
- **Weakened further.** Symptom worsening after orgasm is shown to depend on mechanical stimulation rather than on sympathetic activation.
- **Rendered moot for the closed form.** Any demonstration that sensory discharge in this territory does not modulate pelvic sympathetic outflow.

## 13. Conclusion

Hard flaccid syndrome is discussed as a smooth muscle disorder and experienced, in large part, as a sensory one. The hypothesis examined here — that sympathetic axons sprout within the dorsal root ganglia serving the penis and drive sensory discharge directly — is the only conjecture in this series that addresses why the two should be so tightly coupled, and the only one that explains why three-quarters of patients report worsening with masturbation.

Its foundation is a real, replicated and mechanistically characterised phenomenon, observed in human dorsal root ganglia as well as in rats. Its foundation is also narrower than it looks. The coupling is diffusion-mediated rather than synaptic. The baskets form around the wrong cell population for a pain hypothesis. In a rat study using a sacral ganglion, sprouting magnitude did not distinguish animals with pain from animals without it. The human construct of sympathetically maintained pain has been seriously questioned, and even in its supporting studies only a minority of pre-selected patients respond. The proposed loop closure has no evidence at all.

The right response is neither to adopt the hypothesis nor to discard it, but to split it. Advanced in its open form — as an account of the sensory symptoms and their dependence on sympathetic tone — it is defensible, distinctive and testable by a cutaneous noradrenaline provocation study that requires no new technology and no objective measure of cavernosal tone. Advanced in its closed form, as an explanation of the hard flaccid state itself, it outruns its evidence.

One thing should happen regardless of whether anyone tests this hypothesis. Sympathetic blocks are already being performed in this condition, and their sensory effects are not being recorded. That is a free observation being discarded, and it is the observation that would settle the question.

## Acknowledgements

The underlying conjecture examined in this paper originates with the hard flaccid syndrome community member writing as *Literature Greedy*, and is reproduced with attribution in the source article [9]. We have developed, formalised and critically appraised the idea; we did not originate it. Any errors in its restatement are ours.

## Declarations

**Authorship and the use of artificial intelligence.** The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Every citation, and particularly those marked [VERIFY], should be checked against the primary source first. Attribution of the underlying conjecture to a pseudonymous contributor should also be resolved before submission, ideally by obtaining consent to name or by moving the credit into a formal contributorship statement.

**Ethics approval.** Not applicable. No human participants, animals or identifiable data were involved.

**Funding.** None.

**Competing interests.** No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [9]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

**Data availability.** No datasets were generated or analysed.

**Clinical disclaimer.** This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. The provocation test described has never been performed in genital skin and is not a clinical procedure. Sympathetic nerve blocks,  $\alpha$ -adrenergic antagonists and neuromodulation carry real risk and are appropriate only under qualified medical supervision and, in this condition, only within an approved research protocol.

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